

SYSTEM OF PARTICLES AND ROTATIONAL MOTION

Important formulas and concepts directly from the NCERT-

❖ Centre of mass-

- It is a point where whole mass of body is assumed.
- It can be inside or outside the body.
- It lies near heavy object.

- Displacement of COM,

$$x_{\text{COM}} = m_1x_1 + m_2x_2 \dots / m_1 + m_2 \dots$$

- **Ratio of distance of COM from bodies is inversely proportional to masses of bodies.**

❖ Motion of COM-

- The COM of system moves if net external force is present.
- **If net external force on a system is zero, then COM will not move and momentum of COM will remain conserved.**

$$(F_{\text{net}})_{\text{ext.}} = dP_{\text{COM}}/dt$$

- Kinetic energy of system can also change by internal forces (if external force is zero) because ,

$$W_{\text{by all forces}} = \Delta K$$

But momentum can't be changed by internal forces.

ROTATIONAL MOTION

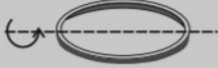
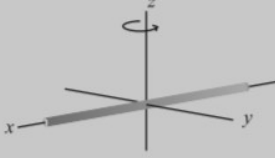
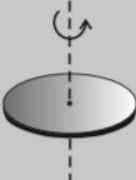
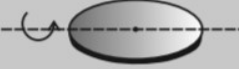
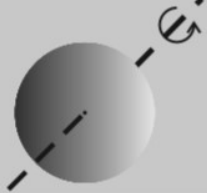
❖ Moment of Inertia (I)-

- Oppose cause of change in rotational state.
- Can be considered as rotational mass.
- Tensor quantity.
- Follow scalar addition.
- **$I = MR^2$ (here, M = mass of object and R = perpendicular distance from axis Of rotation)**

- It depends upon-
- 1) Mass
 - 2) Distribution of mass
 - 3) Axis of rotation

- For two particle system, **I about COM** = $R^2 m_1 m_2 / (m_1 + m_2)$
(R = distance b/w 2 masses)

Table 7.1 Moments of Inertia of some regular shaped bodies about specific axes

Z	Body	Axis	Figure	I
(1)	Thin circular ring, radius R	Perpendicular to plane, at centre		$M R^2$
(2)	Thin circular ring, radius R	Diameter		$M R^2 / 2$
(3)	Thin rod, length L	Perpendicular to rod, at mid point		$M L^2 / 12$
(4)	Circular disc, radius R	Perpendicular to disc at centre		$M R^2 / 2$
(5)	Circular disc, radius R	Diameter		$M R^2 / 4$
(6)	Hollow cylinder, radius R	Axis of cylinder		$M R^2$
(7)	Solid cylinder, radius R	Axis of cylinder		$M R^2 / 2$
(8)	Solid sphere, radius R	Diameter		$2 M R^2 / 5$

9) COM of hollow sphere about diameter = $\frac{2MR^2}{3}$

10) COM of rod about a perpendicular axis passing through one end = $\frac{MR^2}{3}$

❖ Radius of Gyration(k)-

- Distance from axis of rotation where whole mass of body is supposed to be concentrated.

$$I = Mk^2$$

❖ Torque(τ)-

- It is the measure of force that can rotate an object about its axis.
- Also k/as moment of force.
- Vector quantity.
- $\tau = \mathbf{r} \times \mathbf{F} = rF\sin\theta$
- torque depend on 4 factors-

- 1) distance from axis of rotation
- 2) location of axis of rotation
- 3) force
- 4) angle b/w force and distance

- mechanical advantage = large force/small force

❖ rolling motion-

- combination of translational and rotational motion.
- As rod falling from horizontal then its angular acceleration decrease and angular velocity increase.
- Kinetic energy in pure rotational motion,

$$K = \frac{1}{2} I\omega^2$$

- In pure rolling (no slipping condition), velocity of lowermost point is zero and Linear velocity is equal to $R\omega$.
- Total kinetic energy in rolling motion,

$$K = \frac{1}{2} Mv_{\text{com}}^2 [1 + k^2/R^2]$$

➤ $K_{\text{translational}}/K_{\text{total}} = 1/(1+k^2/R^2) = \beta$

Shapes	β
1) Ring or hollow cylinder	0.5
2) Hollow sphere	0.6
3) Disc or solid cylinder	0.67
4) Solid sphere	0.71

➤ in pure rolling on inclined plane,

1) acceleration of com, $a_{\text{com}} = \beta g \sin \theta$

2) time, $t = \sqrt{2H/\beta g \sin^2 \theta}$

H = vertical displacement

3) v_{com} at ground $= \sqrt{\beta 2gH}$

4) maximum height, $H_{\text{max}} = v_{\text{com}}^2/2g\beta$

5) friction force acting,

$$f = mg \sin \theta (1 - \beta)$$

6) minimum coefficient of friction to start pure rolling motion,
 $\mu_{\text{min}} = \tan \theta (1 - \beta)$

❖ angular momentum(L)-

➤ $L = rmv \sin \theta = rP \sin \theta = \mathbf{r} \times \mathbf{P}$ (P=linear momentum)

Or, $L = I\omega$

➤ If net torque is zero, angular momentum is conserved.

$$I\omega = \text{constant}$$

$$I_1\omega_1 = I_2\omega_2$$

- If earth atmosphere is suddenly removed , then I decrease, ω increase and thus time period will decrease.
- Relation b/w L and rotational kinetic energy-

$$K = L^2/2I$$

SUMMARY

1. Ideally, a rigid body is one for which the distances between different particles of the body do not change, even though there are forces on them.
2. A rigid body fixed at one point or along a line can have only rotational motion. A rigid body not fixed in some way can have either pure translational motion or a combination of translational and rotational motions.
3. In rotation about a fixed axis, every particle of the rigid body moves in a circle which lies in a plane perpendicular to the axis and has its centre on the axis. Every Point in the rotating rigid body has the same angular velocity at any instant of time.
4. In pure translation, every particle of the body moves with the same velocity at any instant of time.
5. Angular velocity is a vector. Its magnitude is $\omega = d\theta/dt$ and it is directed along the axis of rotation. For rotation about a fixed axis, this vector ω has a fixed direction.
6. The vector or cross product of two vector $\mathbf{a}$ and $\mathbf{b}$ is a vector written as $\mathbf{a} \times \mathbf{b}$. The magnitude of this vector is $ab\sin\theta$ and its direction is given by the right handed screw or the right hand rule.
7. The linear velocity of a particle of a rigid body rotating about a fixed axis is given by $\mathbf{v} = \omega \times \mathbf{r}$, where $\mathbf{r}$ is the position vector of the particle with respect to an origin along the fixed axis. The relation applies even to more general rotation of a rigid body with one point fixed. In that case $\mathbf{r}$ is the position vector of the particle with respect to the fixed point taken as the origin.
8. The centre of mass of a system of n particles is defined as the point whose position vector is

$$\mathbf{R} = \frac{\sum m_i \mathbf{r}_i}{M}$$

9. Velocity of the centre of mass of a system of particles is given by $\mathbf{V} = \mathbf{P}/M$, where $\mathbf{P}$ is the linear momentum of the system. The centre of mass moves as if all the mass of the system is concentrated at this point and all the external forces act at it. If the total external force on the system is zero, then the total linear momentum of the system is constant.
10. The angular momentum of a system of n particles about the origin is

$$\mathbf{L} = \sum_{i=1}^n \mathbf{r}_i \times \mathbf{p}_i$$

The torque or moment of force on a system of n particles about the origin is

$$\boldsymbol{\tau} = \sum_i \mathbf{r}_i \times \mathbf{F}_i$$

The force $\mathbf{F}_i$ acting on the i^{th} particle includes the external as well as internal forces. Assuming Newton's third law of motion and that forces between any two particles act along the line joining the particles, we can show $\boldsymbol{\tau}_{\text{int}} = \mathbf{0}$ and

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{ext}$$

11. A rigid body is in mechanical equilibrium if

(1) it is in translational equilibrium, i.e., the total external force on it is zero : $\sum \mathbf{F}_i = \mathbf{0}$,
and

(2) it is in rotational equilibrium, i.e. the total external torque on it is zero :

$$\sum \boldsymbol{\tau}_i = \sum \mathbf{r}_i \times \mathbf{F}_i = \mathbf{0} .$$

12. The centre of gravity of an extended body is that point where the total gravitational torque on the body is zero.

13. The moment of inertia of a rigid body about an axis is defined by the formula $I = \sum m_i r_i^2$ where r_i is the perpendicular distance of the i th point of the body from the axis. The

kinetic energy of rotation is $K = \frac{1}{2} I \omega^2$.

POINTS TO PONDER

1. To determine the motion of the centre of mass of a system no knowledge of internal forces of the system is required. For this purpose we need to know only the external forces on the body.
2. Separating the motion of a system of particles as the motion of the centre of mass, (i.e., the translational motion of the system) and motion about (i.e. relative to) the centre of mass of the system is a useful technique in dynamics of a system of particles. One example of this technique is separating the kinetic energy of a system of particles K as the kinetic energy of the system about its centre of mass K' and the kinetic energy of the centre of mass $MV^2/2$,

$$K = K' + MV^2/2$$

3. Newton's Second Law for finite sized bodies (or systems of particles) is based in Newton's Second Law and also Newton's Third Law for particles.
 4. To establish that the time rate of change of the total angular momentum of a system of particles is the total external torque in the system, we need not only Newton's second law for particles, but also Newton's third law with the provision that the forces between any two particles act along the line joining the particles.
 5. The vanishing of the total external force and the vanishing of the total external torque are independent conditions. We can have one without the other. In a couple, total external force is zero, but total torque is non-zero.
 6. The total torque on a system is independent of the origin if the total external force is zero.
 7. The centre of gravity of a body coincides with its centre of mass only if the gravitational field does not vary from one part of the body to the other.
8. The angular momentum $\mathbf{L}$ and the angular velocity $\boldsymbol{\omega}$ are not necessarily parallel vectors. However, for the simpler situations discussed in this chapter when rotation is about a fixed axis which is an axis of symmetry of the rigid body, the relation $\mathbf{L} = I\boldsymbol{\omega}$ holds good, where I is the moment of the inertia of the body about the rotation axis.